

Circuit for leakage current measurement in Power Transistors

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Abstract

The gate leakage current of a power field effect transistor is an indicator of the insulating properties of the gate interface. The leakage current is on the order of nano-amps. Through experimentation, and a circuit which was designed and built, the gate leakage current was measured for four different transistors. The measured current values agreed with the range of values on the respective transistor's datasheets. The built circuit can determine a difference in transistors gate leakage current values and identifying if a transistors gate is faulty.

Acknowledgements

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1 Introduction

A power transistor is a ‘semiconductor device’ designed to regulate current flow and to function as a switch to electronic signals or electronic power (Rabab Khan Rongon, 2022). It can be used in applications such as power amplifiers, switching devices, or a voltage/ current regulator. The aim of the project is to design a circuit for measuring the gate leakage current of a power transistor. In field-effect transistors, the gate leakage current can give an indication of the insulating properties of the gate interface (Claus Fritz Strowitzki, 2013). This can determine if a transistor is faulty or how power efficient the component is and hence its fit for use when manufacturing an electronic device in industry

1.1 Objectives

The project will involve experimental practice involving an extensive design process to create a circuitry device that will output the gate leakage current of a given transistor. There are many different components in the project that will lead to the final product. This includes researching, using software, experimentation in labs, making calculations, budgeting, buying electronic components, microcontroller coding and possibly PCB designing.

The first challenge of the project will be getting a current signal as the gate leakage current on low voltage transistors will have an exceedingly small magnitude, in an order of nano-amps to pico-amps relative to the transistor. The second challenge of the project will be to measure the leakage current over a range of current scales so that the device can be used to measure the leakage current of a variety of transistors. This will require an innovative design method of configuring electronic components thoroughly experimentally and theoretically calculated to be able to take measurements within such an extensive range. The following are the main objectives in order:

- To understand power transistors, specifically MOSFET’s were chosen for the project and the nature of gate leakage current. Research is conducted initially within a feasibility study through personal research supported by published technical reports and existing published research. A thorough understanding of transistors ensures an intuitive approach throughout the project.
- Use software to design of the electronic circuits within the device to be able to help with the physical design. Possible software that can be used may include MATLAB Simscape, Micro-Cap, LTSpice. This will be supported by drawn circuit diagrams and circuit calculations. This helped decide the correct component values and configuration to be able to measure the small magnitude current signal.
- Component selection, using component values form fundamental electronic calculation for component structures. Electronic components are selected referring to relevant published data sheets to be able to identify component compatibility within the circuit. Components are then

ordered online within a £200 budget, using the University of Warwick Engineering School Office order request forms.

- Experimentation, involving building designed circuit using breadboards, resistors, and op amps. Work will be done in labs to configure circuitry to be able to detect a current signal from a given transistor using PSU and DMM equipment. A trial-and-error process will be used involving regular tests to ensure the device works well.
- A micro controller will be coded to be able to output the signal with a digital output useful for data analysis. This will ensure the current is successfully measured and presented coherently. A possible microcontroller to use is a 5V Arduino microcontroller.
- Build a circuit using a PCB board for practical application of device using Altium. This will involve PCB assembly using soldering methods working in labs and engineering build space as well as collaborating with relevant staff in the department to physically create the PCB. This will allow for the possibility of implementing different transistors to be able to measure the gate leakage current of different transistors.
- Collect data of the gate leakage current on different transistors and present the data in a graphical/ coherent way. The entire project will be documented and presented in this report

1.2 Project Duration

The first meeting with the project supervisor Professor Layi was on 1st October 2024 introducing and starting the project. The submission of the project was on 13th March 2025. As project was over a long period, it was important to be effective with time management. To help the project stay on a consistent schedule regular meetings were scheduled with my supervisor to ensure the project stayed on course throughout. A logbook was used to help maintain the development of the project, monitoring budget, progress time management health and safety and wellbeing.

1.4 Budget

Table 1 shows the parts that were ordered and how many of each item was ordered. Not all the purchased components were used in the final design. A research process was taken on each component before purchasing using relevant data sheets, to understand the compatibility with one another. The delivery process took about a week, so time had to be managed to be able to continue with the progression of the project around these delivery times.

The project was permitted a budget of £200 out of this a total of £46.11 was spent. This shows that the budget was managed well as there was plenty of room to buy further parts if needed. The budgeting and time management was managed by using a logbook to track all decisions and timings, throughout the project.

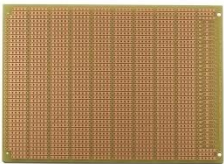
Image	Product	Manufacturing Product Number	Description	Quantity	Price
	Arduino nano (Arduino® Nano., n.d.).	DigiKey- ABX00033	The Arduino Nano Every with Headers (ABX00033) is a compact and versatile evaluation board featuring the ATmega4809 8-bit AVR MCU.	1	\$14.7
	Transistor Holder (DigiKey Electronics.,2025)	DigiKey- TDU03DTOD	3 (Rectangular) Pos Transistor Socket Gold Through Hole	1	\$12.9
	Potentiometer (DigiKey Electronics, 2025).	DigiKey- 450T328F1 03A1A1	10k Ohm 1 Gang Linear Panel Mount Potentiometer 1.0 Turns Carbon 0.5W, 1/2W Solder Lug	1	\$4.17000
	Prototyping Board (BusBoard Prototype Systems -Built for designers, n.d.).	DigiKey- 4526-PR3U-ND	Breadboard, General Purpose Plated Through Hole (PTH) 6 Hole Pad (Both Sides) 0.1" (2.54mm) Grid	2	\$11.23
	LCD display (DigiKey Electronics, 2025).	DigiKey- LCD- S301C31TR	Character Display Module Reflective 7-Segment TN - Twisted Nematic Without Backlight 30.70mm x 16.20mm x 2.80mm	1	\$2.66
	LCD display (I2C TWI LCD1602 Module Gadgeteer Compatible SKU DFR0063., n.d.).	Farnell - DFR0063	Expansion Board, I2C 16x2 Arduino LCD Display Module, Arduino/Genuine UNO/Leonardo Boards	1	£11.86

Table 1:Purchased Components

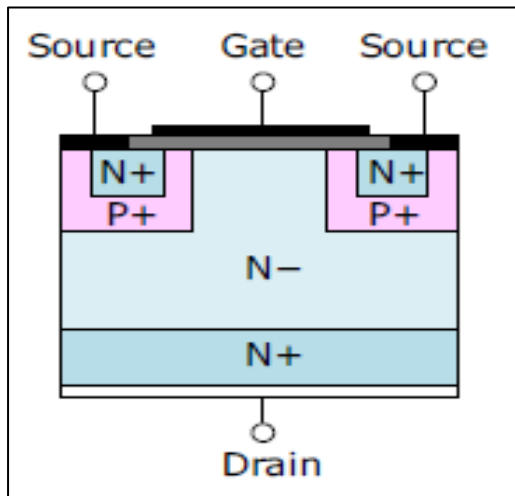
1.5 Risk and Constraints

	Risk on project progress	Effect on Project	Counter-Measure
General project risks			
1	Unable to access required software on personal computer. For example MATLAB and Fusion 360.	Unable to progress on coding parts of project and simulation of circuits quickly.	Request virtual desktop, to be able to access required software remotely.
2	Staying within budget when buying components to build circuits.	Will not be able to buy more components and continue with physical designing part of project	Use simulation, calculations and research with datasheets to ensure correct components are selected.
Time constraints and time management			
3	Ordered equipment has a delayed time in reaching the Engineering department.	Will reduce the time I have to work on physically building the circuit.	Take time to ensure delivery times are sensible and work around these times accordingly.
4	I will be only be able to access the lab at certain times in the week due to the accessibility of the lab to other students.	Time of lab sessions can clash with other commitments through the week or other classes happening in the lab.	Ensure to timetable using logbook when the lab is available to schedule my time in the lab to prevent waiting to work.
5	Making progress too slow within the project.	Will delay future tasks and will result losing time to add possible extra design features.	Use a time line to stick to personal deadlines within the project.
Information, data and security			
6	Unable to save data or work for example code for microcontroller or saved recorded current measurements.	Will limit progress due to inability to work on the task at hand and organise work that has been done.	Work around computer issues by prioritising practical work if there is an issue with wifi or any computer I am using.
7	Loss of data due to saving issues, or potentially faulty or stolen laptop.	Will be detrimental to the progress of my project, and will be hard to work around.	Ensure to use logbook to track project and save files on cloud to reduce risk of lost data
8	Loss of equipment, by theft or potentially misplacing physical designs like circuit boards.	Will be very detrimental to the progress of my project, and will be hard to work around	Ensure to use logbook to track project and maybe use available lockers to store physical work.
Environmental and sustainability			
9	Disposing of electronic components safely and appropriately.	Will not effect my project significantly but may effect the reputation of project.	Use recycling where necessary and donate equipment back to the department if possible.
10	Not keeping labs and School of Engineering Buildingspace clean when doing practical building of circuits and PCB	Can conflict with School of Engineering staff and may have an extensive effect of prohibiting me to use the facilities	Simply ensure to keep the all work stations clean and tidy when working on campus.
11	Negative effect of greenhouse gas emission from shipping brought components from abroad	Will not effect my project significantly but may effect the reputation of project.	Take care to make sure that products are not ordered from too far from the department.
Health and safety			
12	Potential risk of injury from soldering. This can be from the inhaling of fumes from toxic solder melting, or accidentally melting on hands	Injury is not beneficial, it should be avoided during the project. Will effect my ability to build circuits and there is a risk of ruining components.	Wearing suitable PPE for example a mask and gloves to prevent burning my hands or inhaling of toxic fumes, or I can use a desk extractor fan.
13	Burning of equipment by supplying too much voltage to low voltage rated components	Can damage and potentially burn and destroy electrical components.	Ensure to be careful when connecting live wires to components.
14	Potential risk of light shocks from live wires when building circuits.	Not too damaging to personal project but would be ideal to avoid.	Ensure to be careful when connecting live wires to components.
Standards, guidelines and legislation			
15	Lab guidelines must be followed at all times.	Can ultimately result in banning from the facilities.	Read the ESO rules of the lab and ensure to obey by them.
16	Workshop buildingspace guidelines should be followed.	Can ultimately result in banning from the facilities.	Read the ESO rules of the buildingspace and ensure to obey by them.

Table 2: Risk Assessment

2 Literature Review

2.1 Structure of MOSFET



A MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) consists of three main terminals: the Gate, The Drain, and The Source, and is built on a semiconductor substrate. The structure includes two heavily doped semiconductor regions, known as the Source and Drain, which are either N+ doped in an N-channel MOSFET, or P+ doped in a P-channel MOSFET. These regions are embedded in a lightly doped substrate of the opposite type (P-type for N-channel, N-type for P-channel). (Barkhordarian, V, 1996)

Figure 2: MOSFET (semicon.sanken-ele.co.jp., n.d.).

Figure 1 shows a substrate, a thin insulating layer typically silicon dioxide (SiO_2) which separates the Gate from the channel region, preventing direct current flow while allowing an electric field to control conductivity. The Gate electrode is made of polysilicon or metal and is positioned above this oxide layer. When a voltage is applied to the Gate, it influences the charge carriers in the substrate, forming a conductive channel between the Source and Drain. This channel allows current to flow when the MOSFET is switched on, effectively acting as a voltage-controlled switch. (www.mks.com, 2011.).

2.2 Industrial application

The MOSFET was invented in 1959 by Mohamed M. Atalla and Dawon Kahng at Bell Labs. It was a revolutionary development in semiconductor technology, enabling the miniaturization and mass production of electronic circuits. Measuring a MOSFET's gate leakage current is crucial in industrial applications, particularly in power electronics, and more specifically high-frequency circuits. Low gate leakage current ensures minimal power loss and improved reliability, which is essential for energy-efficient devices, circuit stability, and power efficiency. Precise control of the leakage current allows industries to optimise performance given that MOSFETs are mass produced (Armstrong, B., 2023).

The project aims to identify the gate leakage current to identify for use in industry. Erroneous values of gate leakage current suggest a transistor is faulty and hence is unsuitable for use in industry.

2.3 Gate Leakage Current

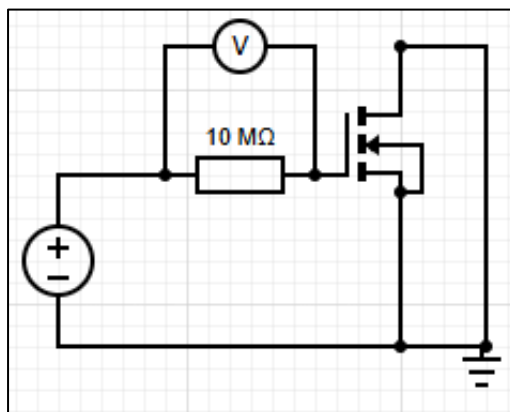
The gate leakage current (I_G) in a MOSFET is the very small undesired current that flows through the gate terminal due to imperfections in the insulating gate oxide layer and quantum tunnelling effects. Quantum tunnelling is a quantum mechanical phenomenon where a

particle passes through a potential energy barrier that it classically shouldn't be able to overcome, due the effect of there being an abundance of charge carries. (J.C. Ranuarez, 2006)

In a theoretical ideal MOSFET, the gate should be completely insulated, allowing no current to flow. Leakage occurs due to oxide thinning, excessively high gate voltage V_{GS} , temperature effects, and manufacturing variations. This means that datasheet stated gate leakage current may not perfectly accurate in some cases hence why a scale or range is often stated.

As MOSFET technology scales down, the gate oxide becomes thinner, increasing tunnel leakage, which can significantly impact power consumption and device reliability. Gate leakage is usually in the range of nanoamps (nA) but depends on the MOSFET design and operating conditions. (None Dongwoo Lee, 2004)

2.4 Method for measuring Gate Leakage Current



The Gate Leakage Current can be measured experimentally by shorting the Drain and Source and supplying a voltage of typical 20V, (so the MOSFET is fully saturated), to the gate of the MOSFET with a shunt resistor series with the gate. The Voltage across the shunt resistor is measured and Ohms Law ($V=IR$). The Measured Voltage is divided by the Shunt resistor value to find the Gate Leakage Current.

Figure 2: Gate Leakage Current Circuit

By shorting the drain and source isolates the gate terminal and, the MOSFET is prevented from conducting normally, ensuring that any current flowing is purely due to leakage through the gate. Applying 20V to the gate creates an electric field across the thin gate oxide. Since the gate is insulated, any current measured is due to quantum tunnelling leakage through the oxide layer only. (TI Designs, 2015)

The gate leakage current measurement circuit works by converting the tiny leakage current into a measurable voltage signal using a precision resistor in series with the gate. When a 20V gate voltage is applied, the small leakage current passing through the resistor creates a voltage drop proportional to the leakage current.

Since gate leakage currents are typically small in magnitude in the nanoamp range a large shunt resistor best suited experimentally in increase the feasibility of measurable the voltage. Given voltage drop is very small and difficult to measure directly, an amplifier can be used to amplify the voltage signal, making it large enough for accurate measurement. This setup ensures that even minuscule leakage currents can be effectively quantified, providing a reliable method for analysing transistor leakage behaviour and minimising the percentage uncertainty in the measurement.

2.5 Drain Leakage Current

Drain leakage current refers to the small amount of unintended current that flows through a transistor's drain terminal when the device is in its off state. Even when a MOSFET, is turned off when the gate voltage is below the threshold voltage to activate conduction), a tiny current, often in the nanoampere or microampere range, can still be observed flowing from the drain to the source or the substrate. (Chen, J., 1986)

This leakage current arises from several mechanisms. One major contributor is subthreshold conduction, where a weak inversion layer forms below the threshold voltage, allowing a limited current to flow, this can be due to thermal effects on the MOSFET like the gate leakage voltage. Another contributing factor is junction leakage, which occurs due to the reverse-biasing of the diode-like junction between the drain and the substrate. Additionally, high electric fields near the drain can lead to gate-induced drain leakage where electrons tunnel through the energy barrier due to the strong field. (Tak Hang Chan, 1987)

Managing drain leakage current is critical in electronic design in industry, particularly in low-power and high-density and frequency circuits. Excessive leakage from off state not only increases power consumption but can also affect overall device performance and reliability. Drain leakage current can give a representation of the Gate leakage Voltage. Drain leakage current is larger in magnitude, so it is easier it measures experimentally.

2.6 Method for measuring Drain Leakage Current

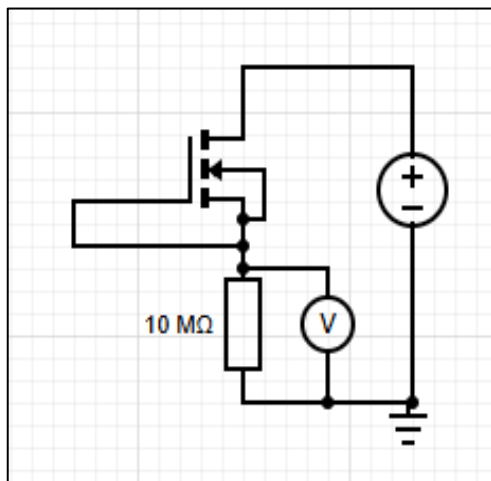


Figure 3: Drain Leakage Current Circuit

The Drain Leakage Current is Measured by shorting the Gate and Source terminals so that the transistor is in off state. and applying a Voltage 20V to the Drain. A Shunt resistor is connected in series with the source terminal. The source (or substrate) is connected to ground. This configuration ensures that the transistor is supposed to be non-conducting, allowing the leakage current to be isolated for measurements. Just like the Gate leakage Current circuit, Ohms Law ($V=IR$) is used to calculate the Leakage current by dividing by the shunt resistor value.

Just like measuring the Gate Leakage current due to the small nature of its magnitude an amplifier can be used to increase the scale of the voltage the shunt resistor. An amplifier used in measurement process also involves careful attention to reducing attenuation in noise and external interference. Using shielded cables, proper grounding techniques, and even controlling unnecessary temperature interferences in the circuit environment can help minimize extraneous signals that might otherwise obscure the true leakage current. This

ensures that the readings are accurate and repeatable, which is critical in low-power and high-precision applications in industry. (Low Current Measurements., n.d.).

2.7 Potential Divider Circuit

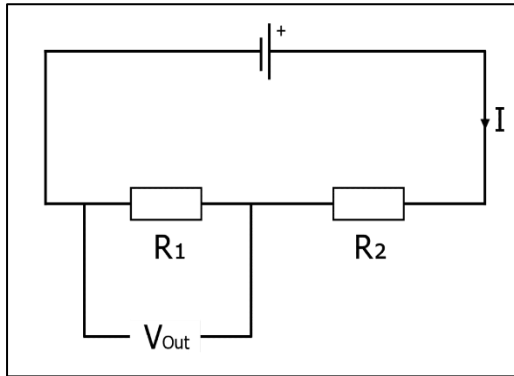


Figure 4: Potential Divider (MME., n.d.)

A potential divider, also known as a voltage divider, is based on fundamental electrical laws—primarily Ohm's Law and Kirchhoff's Voltage Law (KVL). In a simple series circuit consisting of two resistors, Ohm's Law ($V = IR$) tells us how voltage drops across each resistor are proportional to their resistances, while KVL ensures that the sum of these voltage drops equals the total supplied voltage. (MME., n.d)

$$V_{out} = V_{in} \left(\frac{R_1}{R_1 + R_2} \right) \quad (1)$$

Here, the output voltage (V_o) is a fraction of the input voltage (V_i) determined by the ratio of resistor R_1 to the sum of both resistors.

2.8 Noninverting amplifier

A non-inverting amplifier is a commonly used operational amplifier (op-amp) configuration. The input signal is applied to the non-inverting positive input terminal (+), and a feedback loop is connected to the inverting negative terminal (-). The setup allows the amplifier to produce an output voltage that is in phase with the input, making it ideal for applications where phase preservation is necessary in signal output. The circuit consists of a feedback resistor (R_2) and a gain resistor (R_1), which together determine the voltage gain. The gain of a non-inverting amplifier is given by the formula:

$$A_v = 1 + \frac{R_2}{R_1} \quad (2)$$

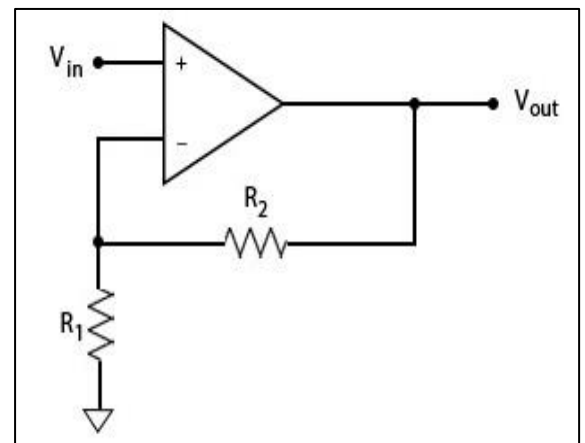


Figure 5: Non-Inverting Amplifier (Analog Devices, 2024)

Where (A_v) is the linear voltage gain, (R_2) is the feedback resistor, and (R_1) is the resistor connected to the ground. The equation shows that the gain is always greater than or equal to 1, so the circuit can only amplify or buffer the signal but never attenuate it. The non-inverting amplifier has high input impedance, which minimises loading effects on the source, and low output impedance, which allows efficient signal transfer. Due to this, it's commonly used in

audio amplification, sensor signal conditioning, active filters, and buffer circuits. (Khateb, F, 2015)

2.9 Buffer

A buffer is a circuit designed to isolate different parts of a system, preventing unwanted interactions and ensuring signal integrity. The buffer can be configured by removing the ground resistor in a non-inverting op amp and adding a wire feedback from the output voltage terminal to the input negative inverting terminal. Using the non-inverting amplifier equation (1) this gives a linear gain of 1 (Unity Gain). Given R_1 and R_2 are zero.

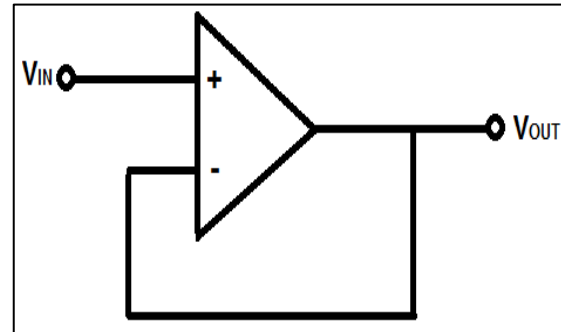


Figure 6: Buffer (Sstatic.net, 2025).

The Buffer acts as an intermediary, maintaining the input signal while providing sufficient power, voltage, or current to drive the next stage. Buffers are crucial in digital and analogue circuits to prevent loading effects and stabilize signals by reducing noise. They are commonly used in microcontrollers, signal processing, and communication systems to ensure reliable and distortion-free transmission.

The Structure of the buffer incorporates a negative feedback loop structure, where a portion of the output signal is fed back to the input in an inverse manner, helping to stabilize the system and reduce distortion. In the case of an op-amp voltage follower (unity-gain buffer), the output is directly connected to the inverting input, ensuring that the op-amp always adjusts itself to maintain the input voltage at the same level as the output. This results in high input impedance, low output impedance, and unity gain, making it ideal for isolating and driving subsequent circuit stages without signal degradation, this can help when measuring small voltages with a large resistive load.

2.10 Threshold voltage

The threshold voltage (V_{th}) of a MOSFET is the minimum gate-to-source voltage (V_{gs}) required to induce a conductive channel between the drain and source terminals. In an n-channel MOSFET, the device remains off when V_{gs} is below V_{th} , but once it exceeds this threshold, an inversion layer forms within the channel, allowing current to flow. For a p-channel MOSFET, the threshold voltage is negative, and conduction occurs when V_{gs} is below V_{th} . The exact value of V_{th} depends on factors like doping concentration, oxide thickness, body bias, and temperature of the semiconductor substrate. Typically, lower V_{th} values are used in high-speed and low-power applications and MOSFETs with higher thresholds can handle larger voltages and currents more efficiently. (Terada, K, 2001)

2.11 Method for measuring threshold voltage

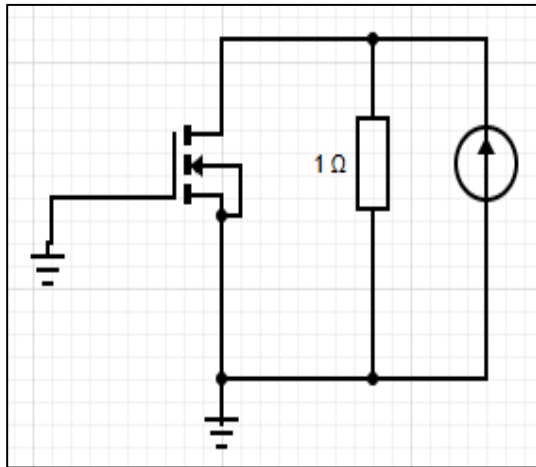


Figure 7: Threshold Voltage Circuit

The Threshold Voltage is measured by a circuit configured in a source follower (common source) arrangement with its drain connected to a constant current source and a resistor placed in parallel with the source path. When the MOSFET begins to conduct, the voltage drops across the resistor corresponds to the source current. Since the gate is grounded, the source voltage (V_S) is directly equal to the threshold voltage (V_{TH}) of the MOSFET. By measuring the voltage at the source terminal, the threshold voltage can be determined directly.

(and, E., 2023).

Similarly, the threshold voltage can be determined in simpler method. With a current supplied to the Drain terminal by any resistor in source with a voltage source and the Source terminal connected to ground. Another power supply source can be used to supply the gate voltage, this voltage on an N type MOSFET can be increased slowly from zero and the voltage across the drain resistors can be noted to determine what threshold voltage allows current to flow. The current through the drain resistor can be plotted against the Gate Source voltage (V_{GS}) to better determine the Threshold Voltage.

The Threshold Voltage circuit was used in the project to help understand the reliability of the MOSFET and to see if it was functioning normally at certain steps of the design process. The Threshold voltage circuit was not used in the final circuit design with the current source.

2.12 Relevant Literature

The methodology for measuring the transistor properties was understood from the project supervisor. However, an exhaustive research process was taken to understand present methods for measuring transistor gate leakage current. This was done through the means of published research papers.

In the paper ‘*Monitoring of Gate Leakage Current on SiC Power MOSFET*’, the authors propose an indirect method to estimate gate leakage current using a smart gate driver rather than direct measurement, which is challenging due to the low amplitude of the leakage current. A gate driver is a circuit that controls the switching of power semiconductor devices like SiC MOSFETs, ensuring efficient operation and protection from faults. The proposed estimation method utilizes the capacitive behaviour of the gate-source interface by monitoring the source voltage (V_S) variation over time when the static bias voltage stage is disconnected. By observing the time (TL) required for the source voltage to change within a predefined range (ΔV_S). (Weckbrodt et al., 2021).

The authors explain how gate leakage current can be indirectly estimated using a mathematical model that accounts for circuit parameters such as gate capacitance and resistance. This method was validated experimentally by comparing results with aged SiC MOSFETs, showing that leakage currents as low as 50 nA could be detected, making it a viable solution for real-time health monitoring of power transistors in energy conversion systems (Weckbrodt et al., 2021). While using gate drives are not within the scope of the project, due to facility limitations inspiration is taken from the ethos of the verification of the experiment by using various aged transistors with different gate leakage current values within the nanoamp range.

The paper "Ultra-low noise high electron mobility transistors for high-impedance and low-frequency deep cryogenic readout electronics" focuses on the development and characterization of high electron mobility transistors (HEMTs) designed for operation at deep cryogenic temperatures (4.2 K). These transistors exhibit ultra-low noise levels and are intended for applications such as dark matter detection and quantum experiments. (Dong, Q, 2014)

The study includes an experiment to measure gate leakage current, where a specially designed circuit was used to determine the leakage current of the transistor. The experimental setup involved a capacitor-based input configuration, where the voltage variation across a capacitor was monitored to infer the leakage current. The results showed an extremely low gate leakage current of 0.62 attoamperes (aA), which significantly reduces low-frequency noise. (Dong, Q, 2014)

The paper highlights that high impedance is critical for maintaining low noise performance, and reducing gate leakage current is essential for achieving high-impedance operation. The authors note that in high-impedance circuits, even tiny leakage currents can degrade performance by introducing excess noise (Dong, Q, 2014). Using a capacitive based setup and instantaneous measurements with machines cannot be achieved also due to facility limitations within this project. However, understanding the importance of using a high impedance to measure gate leakage current to minimise voltage noise is useful for taking measurements experimentally. This was learnt through the experimentation process after when a shunt resistor used to measure gate leakage current was measurable with a visible with unity gain buffer.

3 Experimentation

3.1 Experimentation process

The entire experimentation process involved using scheduled lab in the EPL in the Warwick School of Engineering. The purpose of the lab sessions was to initially understand how to use the lab equipment and to understand the nature of the MOSFET experimentally. Later lab sessions were used to design a sensor and code a microcontroller using hardware to ultimately measure the gate leakage voltage of different MOSFETs using a bread board design. It is important to note from the outset that the theoretical values provided in datasheets of MOSFETs, resistors and all used electronic components do not perfectly align with the experimental measurements due to inherent uncertainties in manufacturing of components and measurements in the setup.

One of the primary sources of discrepancy in measurements is the tolerance in resistor values. Typically, resistors come with a 1 to 2 percent uncertainty, meaning the actual resistance may differ slightly from the stated value. This variance can influence key parameters such as the threshold voltage and gate leakage current specified in the MOSFET datasheet, as well as various parameters on the op-amp datasheet.

Furthermore, the gate leakage voltage cannot be measured directly with a DMM because its magnitude is too low to be accurately captured. Instead, an indirect measurement method is employed by using a sensor in the form of an amplifier. The entire measurement process was taken indicatively so any issues with the amplifier, lead to erroneous readings could be diagnosed. To mitigate this risk, a sensor will be used to monitor the drain leakage current as a reference as the Drain leakage current was easily measured without the sensor due to it being larger in magnitude. This approach helped to ensure the reliability of the setup and confirm that the observed gain closely approximates the theoretical predictions before proceeding further with the experiment.

3.2 Chronological method

It is important to understand that there can be many issues with the design. This can be from the code (syntax or logically) of the microcontroller, unstable cable connections, faulty parts (possibly from damage from too large voltages), or incorrect configuration of circuitry. Given that there are many variables that can form an issue in the circuitry an extensive design process is used taking small steps to add small additions to get the circuit closer to being able to complete its function. This is because an immediate initial design will have a higher probability of an issue than at each individual step. Similarly, it would be very difficult to diagnose what is wrong with the design if too many steps are taken at once

During the process this slow extensive process is taken. The first couple of lab sessions were used to understand the rule so the lab as well as an insight of the availability and booking process to use it and the health and safety rules and regulations. Also, time was taken to understand and gain a familiarity with the equipment, simple test of measuring voltage across resistors with a multi-meter using a power supply voltage.

The experimental process starts by unsuccessfully measuring gate leakage current due to the small size of it and but successfully measuring the drain leakage current due to its larger scale. A sensor is made then to measure the drain leakage current to test the sensors validity by comparing with the multi-meter measurement. After validating the sensor is used to find the gate leakage current. At each step the sensor is tested first with a unity gain test for similar results then with gain to test for amplified results. This process is logical and follows an approach that ensures accuracy in the final measurement.

3.3 Equipment

The final goal is to create a circuit to be able to measure the gate leakage voltage with a microcontroller. While a microcontroller's signal readings are limited due to the digital quantised limitations of measurements, the fast reading and easy data manipulation allows for easy analysis of results. Initially a digital multimeter (DMM) was used to measure the gate leakage voltage and only other intermediate potential differences, ensuring it is within 0 –5V to ensure its output is compatible with a microcontroller. As this cannot be directly measured a circuitry must be designed through experimentation to be able to measure the gate leakage voltage.

The DMM that was used during experimentation was the 'AMPROBE AM-510'. From reviewing the datasheet, it can measure voltage up to 40V and has a resolution of 10mV. As current readings are not that accurate the DMM is not used for that. This was also used to ensure the accuracy of resistor values by using the resistor setting on the DMM. (VolTect™, 2012)

During experimentation two different power sources were used. Primary the Keithly '2231A-30-3 triple channel DC Power Supply' was used to power amplifiers used in designing sensors for the measurements. The PSU has 3 terminals, two can provide up to 30V with a constant current and 1 can provide up to 5V, with a constant current. (A Greater Measure of Confidence., 2014). Due to the limitation of the third channel on the Keithley PSU the BK Precision model '1550' was also used to supply 20V the MOSFET. (B&K Precision Corp, 2014),

The voltage supply for the amplifier requires a positive and negative supply of rail voltages, in the circuitry in experimentation and the final circuit. This is achieved by connecting two PSU terminals in series by connecting one ground to another positive source. The ground of one power supply is the negative terminal and the positive power probe of the other is the positive power terminal. This is achieved by combining the grounds to a reference ground.

The MOSFETs used in the experimentation process had a distance between the pins of 2.54mm which is compatible with a bread boards and strip boards so is suitable for testing and replacing easily, The MOSFETS used are IRL530BPF (IRL530 Power MOSFET, 2021), which has a data sheet and MOAFET code ‘nxp op629 pem0932 b3 9865’ manufactured by NXP Semiconductors as a prototype which is intended to have a slightly larger gate leakage current, which doesn’t have a data sheet.

The final circuit used larger transistors with a pin spacing of 10.16 mm. These transistors pins are too large and far apart to be inserted into the bread board and hence used for experimentation. Wires were used to connect the pins the bread board however, unstable conductive connections related to unreliable results. Hence, smaller transistors with 2.54mm pin spacing, compatible with a standard bread board was used for experimentation.

3.4 Gate leakage current measurement observation

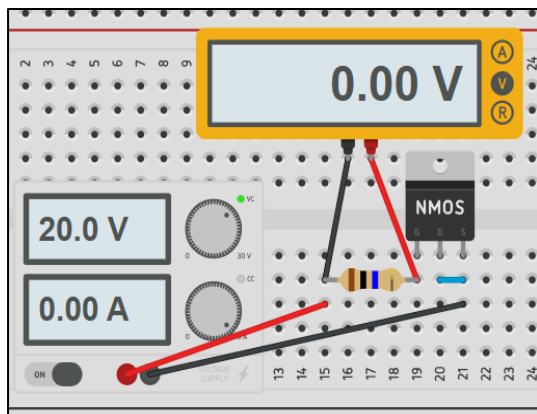


Figure 8: Experiment 1 model

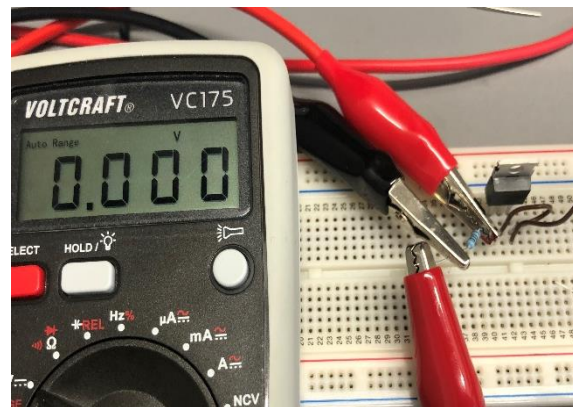


Figure 9: Experiment 1 Bread Board

Initially an experiment was taken to directly measure the gate leakage current of the IRL530PBF MOSFET. The drain and source are shorted on the bread board and there is a shunt resistor connected to the gate. 20V is supplied to the shunt resistor and ground is connected to the drain source connection. A DMM is used to measure the voltage across the shunt resistor. Initially there is a voltage of zero across the shunt resistor of 10K ohms. The shunt resistor is increased to 10M ohms and there was still no voltage across the resistor. This means that the induced gate leakage currents smaller than the smallest scale of the DMM. The Resistance was increased to 100M ohms by combining resistors in series but there was still no voltage signal.

If the gate leakage current is in the nano amp range using a Mega ohm resistor should give a voltage in the millivolt range. As there was no voltage, it's unclear to determine is a lack of signal is due to the current being exceedingly smaller in the pico-amp range or due to other factors, like impedance loading issues.

When using the 10M ohm resistors a voltage signal was perceived when two fingers were used to pinch the resistor. The voltage on the DMM was perceived as an impulse value exponentially reached. The reason for this observation can be due to the increase in resistor value because of an increase in thermal energy of the resistor. This means that the voltage across the shunt resistor to determine the gate leakage current can be measured, but a method is required to be able to read the voltage. A sensor is thus required to amplify the voltage of the shunt resistor in series with the MOSFETs gate.

3.5 Sensor

Given the gate leakage current is very small in scale, utilising a large resistance is required to be to measure the voltage across it with the DMM to be able to calculate the gate leakage current. Increasing this voltage can be done by increasing the gate resistor by adding resistors in series however this does not increase the factor of the measurement, so a voltage amplifier is required to be able to measure this voltage with the DMM, which also means measuring with a greater degree of accuracy by reducing the uncertainty of the measurement.

An operational amplifier was designed to amplify voltages. OP177GPZ (2b, C., 2016.) was initially used to design a unity gain amplifier. Using the information from the data sheet -15V and $+15\text{V}$ was applied to pin 4 and 7 respectively to supply power to the amplifier, from *figure 11*. A wire with no resistance is used as a feedback loop from pin 6 the output pin to pin 2 the negative input signal pin. A separate source on the power supply is used to input a signal voltage, the positive terminal is connected to pin 3 the positive input voltage terminal, and the ground of the signal with respect to its voltage is connected to the end of an output resistor of 100Kohms.

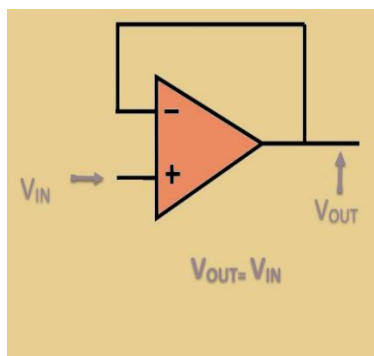


Figure 10: Buffer diagram

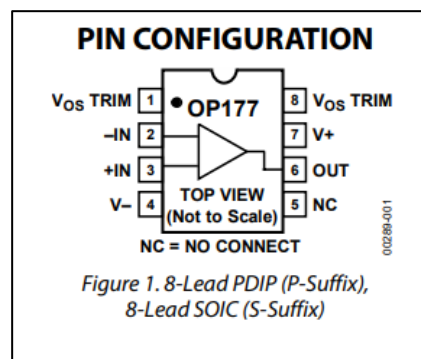


Figure 11: OP177GPZ pin configuration

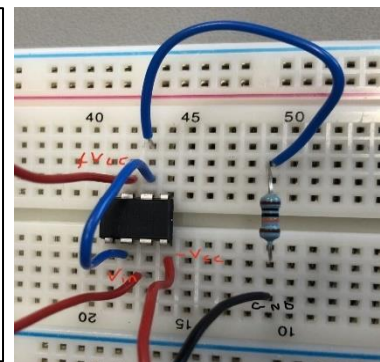


Figure 12: Buffer on breadboard

The potential difference of the resistor is measured with a DMM to determine the validity of the operational amplifier. When voltages of 1V to 5V in increments of 1V are input to the amplifier, the same voltage was measured using the DMM within an error of -10mV . The output voltage was consistently below the input voltage. This means that OP amp correctly provides linear unity gain (linear gain of 1) but is minimally reduced due to internal resistance within the OP amp

The Gain of the Amplifier can be determined using equation (2), where the reference and ground resistors are equal to zero hence substituting into the equation gives a linear gain of 1. This can be determined experimentally as visualized from *figure 13* where the gradient of the graph represents the linear gain which is approximately 1.

Input PSU (V)	1V	2V	3V	4V	5V
Output DMM (V)	0.999V	1.999V	2.999V	3.998V	4.98V

Table 3: Unity Gain Input Output Voltages

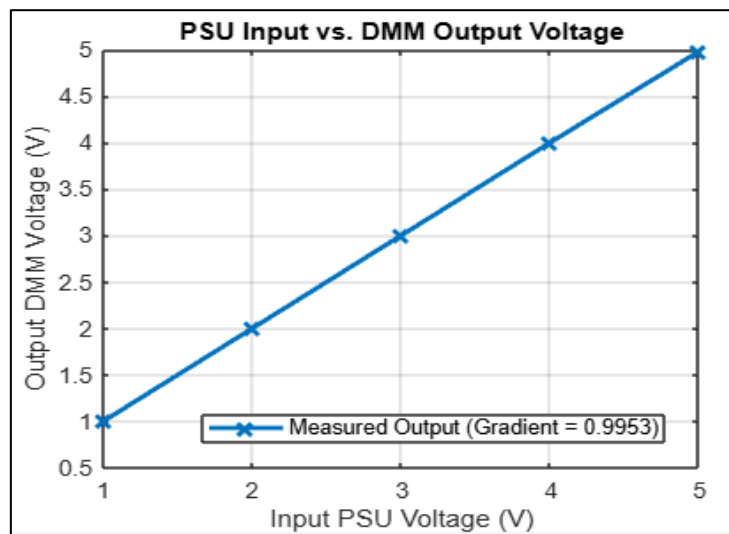


Figure 13: Linear Unity Gain Graph

A non-inverting amplifier was then formed from the unity gain design to create a linear voltage gain. This used the same setup as the unity amplifier powered by +15V connected to pin 7 and -15V connected to pin 4 from two channels from the PSU and another channels positive terminal connected to pin 3 the positive input signal voltage and its ground terminal of the channel connected to the output resistor. The negative feedback loop wire was replaced with a feedback resistor (R2 in *figure 5*) which connected from pin 6 the output signal voltage terminal to pin 2 the negative signal voltage input terminal. An addition ground resistor (R1 in *figure 5*) was connected from pin 2 to ground which is in series with the output resistor and ground of the PSU input voltage channel.

The resistor values were determined using the non-inverting amplifier configuration equation (2). Given the ratio of reference resistor to the resistor connected to ground is directly proportional to the gain. Initially a gain of 11 is designed with a 10Kohm resistor used as R2 connected from the negative signal input pin 3 and ground and a 100kohm resistor connected as a negative feedback loop between pin 8 and pin 2. The output resistor of the amplifier was kept the same as the feedback resistor of 100k ohms.

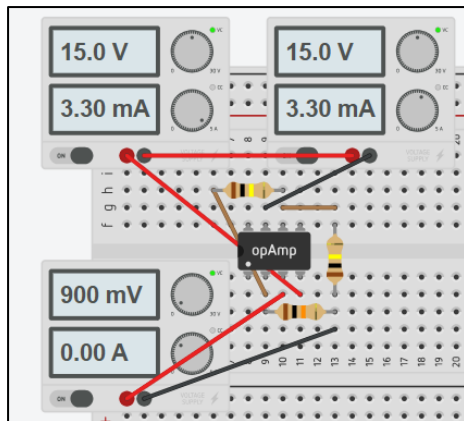


Figure 13: Amplifier Model

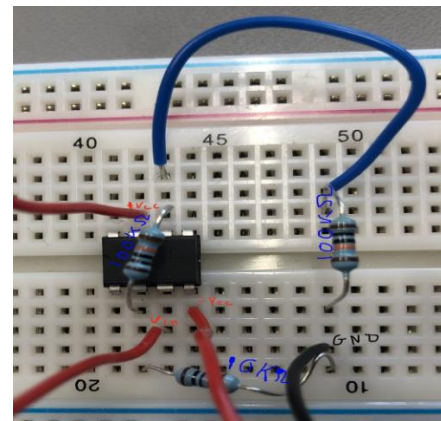


Figure 14: Bread Bord Amplifier

Input PSU (V)	0.01V	0.05V	0.1V	0.5V	1.0V	1.1V	1.2V	1.3V	1.4V	2-5V
Output DMM (V)	0.108V	0.55V	1.02V	5.50V	11.02V	12.12V	13.22V	14.27V	14.27V	14.27V

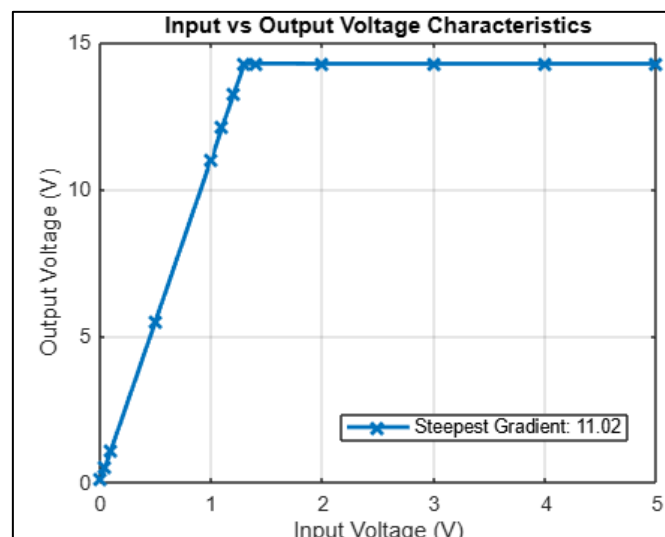
Table 4: Amplifier ($G=11$) Input Output Voltages

Figure 11: Amplified Gain 11 Graph

From the designed amplifier circuit, the DMM was used to measure the output voltage across the resistor in series with the output pin and the input voltage from the third channel, its input signal voltage, of the PSU was noted. The plot of these voltages *figure 11* shows there is a linear gain of 11 from the gradient of the linear increasing part of the graph. At about 1.3V the output voltage stops increasing and remains at 14.27V. This is due to the limit of the power rails used to power the operational amplifier of 15V. The limited voltage is likely below the source voltages due to either internal resistance within the operational amplifier and the resistors not being the exact values as stated. The sensor successfully provides voltage gain but for only small voltages.

The Amplifier circuit was built again, designed with a linear gain of 101. This was achieved by replacing the feedback and output resistor with a 1M ohm resistor. Using equation (2) the non-inverting amplifier gain is 101 due to the ratio of the feedback resistor to the ground resistor. When The Keily PSU supplied an input voltage of 0.01V to the amplifier 9.93V measured by the DMM giving an approximate gain of 101. There is a greater uncertainty in the experimental output voltage relative to the input gain due to a significantly larger gain used on a smaller voltage.

When 0.02V was input to the amplifier 14.34V measured by the DMMs output voltage which aligns with what is expected due to its expected amplified voltage being greater than the power terminal voltage of the op amp. With only one measured value it was not possible to plot the results or determine the reliability of the amplifier with a gain of 101.

3.6 Microcontroller

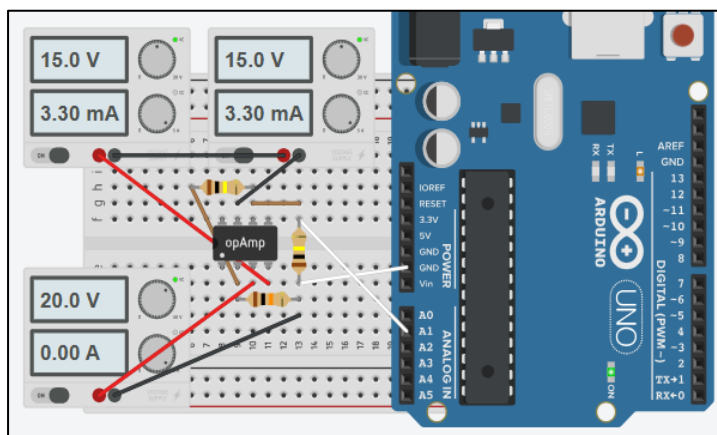


Figure 12: Microcontroller Sensor Model

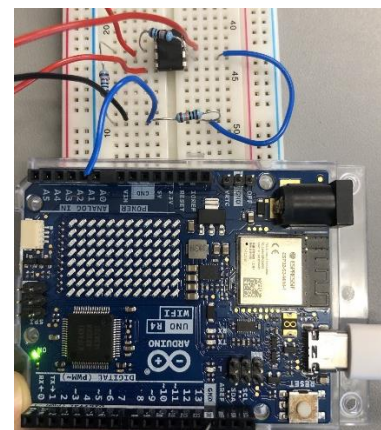


Figure 13: Microcontroller Sensor Breadboard

The Arduino Microcontroller was initially used to replace the DMM to in the voltage sensor. The Arduino used for experimentation was the 'Arduino Uno R4 WIFI' (Arduino Official Store, 2023). Similarly, how the DMM measured the voltage across the output resistor of the amplifier the Arduino measured the voltage across the output resistor. From the amplifier circuit with a gain of 11 the ground of the output resistor was connected to the ground pin on the Arduino and the other side of the resistor was connected to analogue pin 1 on the microcontroller.

The ATmega controllers used for the Arduino contain an onboard 6 channels for analogy-to-digital (A/D) conversion. The converter has a 10-bit resolution, which returns integers from 0 to 1023 using the 'analogRead' command. (Arduino.cc. (2024). Given that The Arduino can only read an input up to 5V which is represented by a digital value of 1023. The Output voltage of the sensor is calculated by dividing the analogue value by 1023 and multiplying by 5 to get the voltage. This is done by the following code:

```

void setup() {
    Serial.begin(9600); // Start serial communication at 9600 baud
}

void loop() {
    int analogValue = analogRead(A1); // Read the analog value from A1
    float voltage = analogValue * (5.0 / 1023.0); // Convert to voltage (assuming
5V reference)

    Serial.print("Analog Value: ");
    Serial.print(analogValue);
    Serial.print(" | Voltage: ");
    Serial.print(voltage, 3); // Print voltage with 3 decimal places
    Serial.println(" V");

    delay(500); // Delay for readability (500ms)
}

```

Input V	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
DMM V	1.103	2.207	3.312	4.400	5.510	6.610	7.710	8.810	9.920
Arduino V	1.144	2.104	3.151	4.156	4.813	4.934	5.000	5.000	5.000

Table 5: Amplifier (G=11) DMM and Arduino Voltages

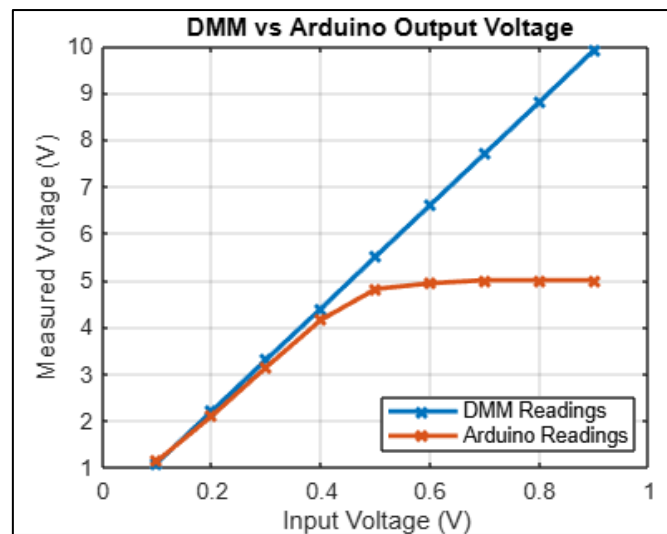


Figure 13: Amplified Gain 11 Microcontroller Graph

When the output voltage of the amplifier is measured it is clear that for small voltages that microcontroller measured values aligns with the DMM measured value of output voltage of the amplifier. There is an uncertainty of $\pm 0.1V$ due to the limitation of the quantisation error of the digital readings in the microcontroller. As the input voltage increases to about 0.4V the Arduino measured voltage plateaus at 5V due to the 5V limit of analogue readings on the microcontroller. This is like the limitation of 15V due to the power supplied to the op amp when measured from the DMM. This limitation was taken into consideration to ensure outputted voltage were below 5V.

3.7 Experimentally Measuring Drain Leakage Voltage

The NXP MOSFET was used to measure the drain leakage current. The MOSFET has no online documentation, but the gate drain and source terminal locations are known, and it is known the threshold voltage is between 0V and 2V. The gate and Source are shorted so that there is no potential difference across the terminals and the MOSFET is in an off state. A shunt resistor is connected in series with the source terminal and 20V is supplied using the Keilby power supply to the drain, with the ground connected to the shunt resistor. The DMM is used to measure the voltage across the shunt resistor and the known resistance using Ohm's law is used to find the drain leakage current by dividing the voltage by the shunt resistor value.

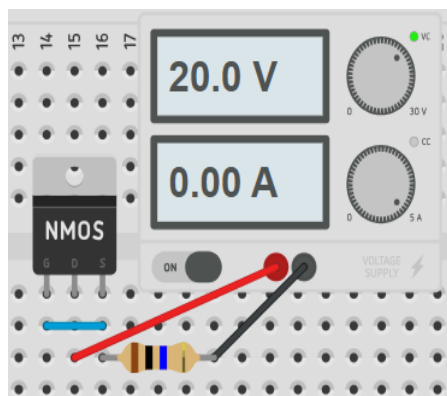


Figure 14: Drain Leakage Current Model

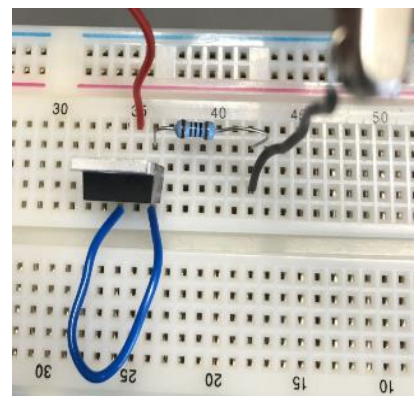


Figure 15: Drain Leakage Current Bread Board

The Shunt Resistor was increased in value to be able to see an output signal voltage across the shunt resistor. With a small resistor (any resistor below 10M ohms), the voltage across the resistor is too small to be measured with the DMM. 0.001V is measured with 10M resulting in a 1nA current. Increasing the shunt resistor by adding resistors in series decreases the current but the current remains in the nano amp scale. The linear increase in shunt resistance doesn't linearly decrease the current. This is because the Drain Source resistance measured with the DMM is initially 10.72M ohms. This means that current is not representing ohmic conditions due to a potential divider effect from the drain source resistance and the shunt resistor. Also, there may be variation in this resistance which was not measured to be a reason for a decrease in current with increase in current.

Shunt Resistor	1kohms	10kohms	10Mohms	20Mohms	30Mohms	40Mohms
Resistor voltage	0.00V	0.00V	0.001V	0.017V	0.019V	0.018V
Current (Ids)	0nA	0nA	1nA	0.75nA	0.63nA	0.45nA

Table 6: Drain Leakage Current Varying Shunt Re

From the setup of the drain leakage current measurement a test is done to verify the measurements are valid by seeing how the current across the drain resistor varies with varying the gate source voltage. As the gate voltage is increased from 0 to 3V *figure 18* an

exponential increase is seen. It seems like the threshold voltage from this is about 2.6V as that is when current can flow and the MOSFET is fully saturated. This observation is as expected so the drain leakage current measurements can be validated.

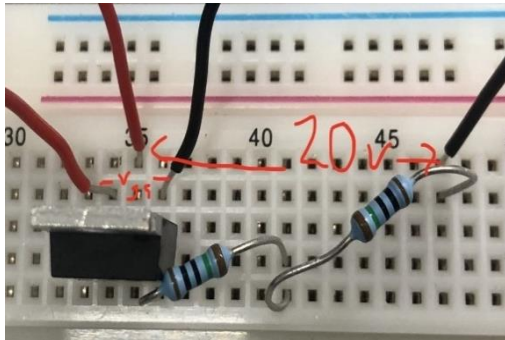


Figure 16: Varying V_{gs} Breadboard

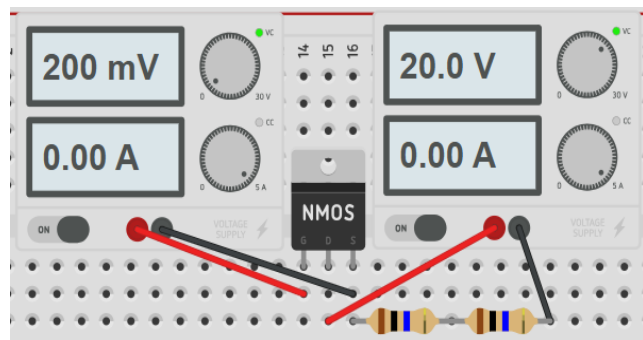


Figure 17: Varying V_{gs} Model

Shunt Resistor V	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.015	0.019	0.031	0.078	0.277	1.105	3.94
V_{gs}	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0
I_{ds}	70nA	70nA	70nA	70nA	70nA	70nA	70nA	70nA	75nA	95nA	1.55nA	3.9nA	0.139nA	0.553nA	0.00197nA

Table 7: V_{gs} against I_{ds} plot

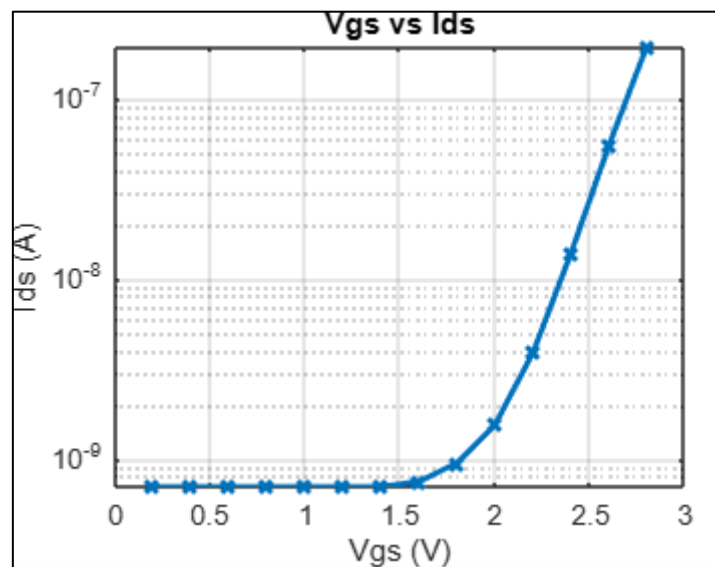


Figure 18: V_{gs} Against I_{ds} Plot

3.8 Amplifying Drain Leakage Current

Given the drain leakage current can be measured with the DMM the sensor is used to measure the leakage current of the MOSFET. This is done to test the validity of the sensor given the drain leakage current is known for different source series shunt resistor values. Initially the unity gain amplifier configuration is connected to shunt resistor. The electrical potential at the point after the gate source short circuit is connected to the positive input

signal voltage of the op amp and the potential after the resistor is connected to the ground of the op amp which is after the output resistor.

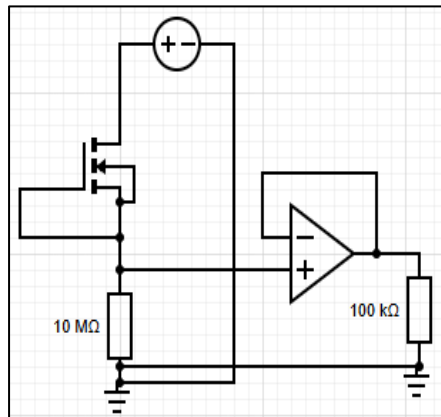


Figure 19: Ids Unity Gain Diagram

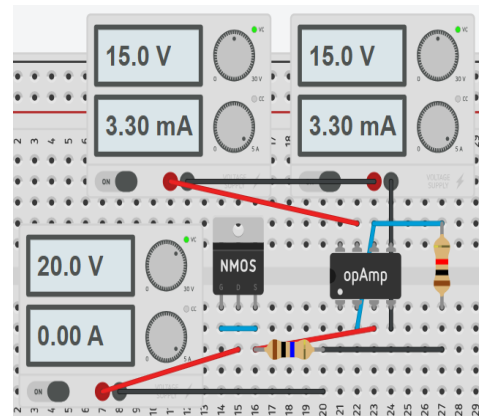


Figure 20: Ids Unity Gain Model

The Resistor voltage in series with the drain leakage current is increased from 10M to 20M to 30Mohms. With the 10M ohm resistor the amplified voltage seems to correlate with the voltage across the resistor in series with the source voltage. Incrementally increasing the source resistance has the same non ohmic effect on the increase in resistor voltage due to the nature of the potential divider due to the internal Drain Source resistance of the MOSFET. When increasing the resistance, the amplified voltage increasing in an ohmic way to the increased resistance.

The reason for the unity amplified voltage not being equivalent to the source resistor voltage, is not fully understood. Possibly the buffer outputting a signal with a large impedance can stabilise the drain source voltage so ensuring that the output voltage of the source resistor increases proportional to the resistor increase. Regardless of this observation, this is taken in regard when repeating this process with gate leakage current.

Source Resistor	10M ohm	20 M ohm	30 M ohm
Source Resistor V	0.009V	0.013V	0.014V
Amplified Resistor V	0.009V	0.020V	0.034V
Vds	9.99V	6.67V	5.00V

Table 8: Unity Amplifier Drain Leakage Voltage

The experiment is repeated with the designed sensor with the gate and source shorted and 20V applies the drain with an initial shunt resistor of 10M ohms connected in series with the source after the short circuit node. An amplifier is connected to the shunt resistor with a gain of 11.

To configuration of the circuit was ensured to be correct before taking any measurements. It was important to connect the ground of the op amp power rails from the PSU to the ground terminal of the ground resistor on the non-inverting op amp and the zero volt of the output

resistor and the ground of the shunt resistor. This ensures that all ground on the PSU, the MOSFET configuration and the amplifier are all aligned so there are no imaginary grounds so that the measured potential difference is correct relative to the circuits ground.

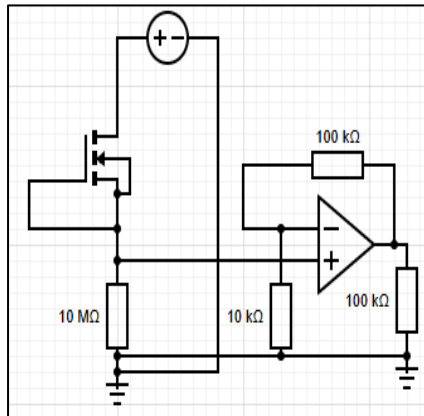


Figure 21: Ids Amplified Diagram

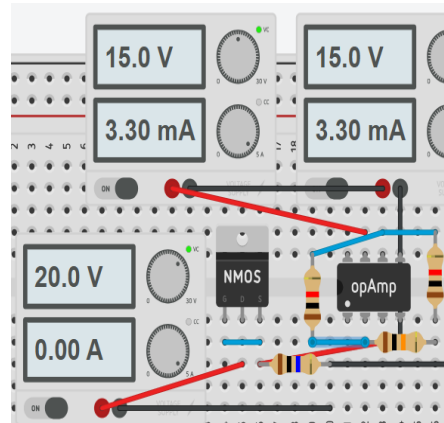


Figure 22: Ids Amplified Model

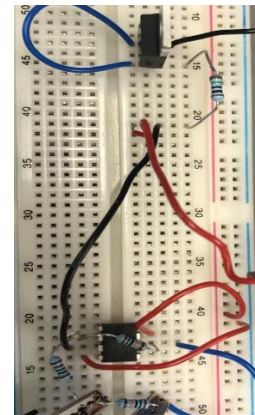


Figure 23: Ids Amplified

The test has the same results as before, with the source resistor voltage, exhibiting the same non ohmic effects when its resistance is increased from 10M to 20M to 30M ohms. The amplified voltages again don't align with the measured DMM source resistor voltage but seems to align with an amplification of 11 to the output voltage of the unity gain amplifier to the amplifier with a gain of 11. The same process was repeated with a gain of 101 by using a 1Mohm resistor as the output resistor and the feedback resistor in the amplifier. Conclusively a 10Mohm resistor is used as the shunt resistor when measuring gate leakage current and the gain sensor is valid to use for measuring transistor voltages.

Source Resistor	10M ohms	20M ohms	30M ohms
Resistor voltage	0.009V	0.013	0.014V
Amplified V	0.117V	0.227V	0.371V
Vds	17.69V	17.69V	17.3V

Table 9: Amplified Drain Leakage Voltage

3.9 Amplifying Gate Leakage Current

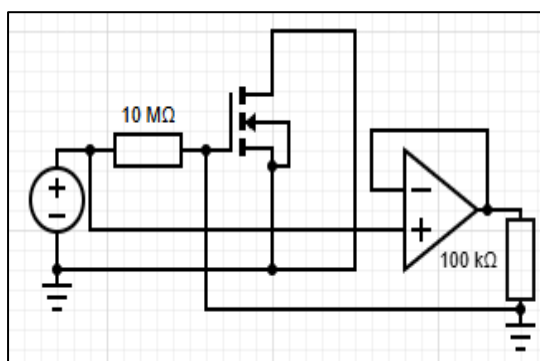


Figure 24: Ig Unity Amplified Diagram

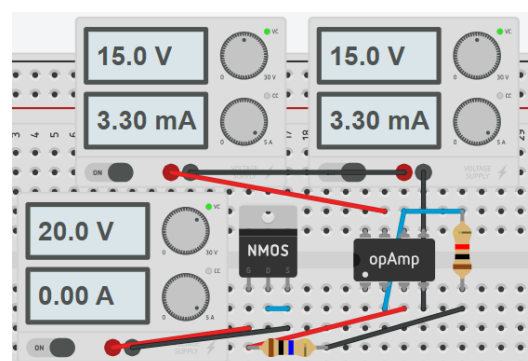
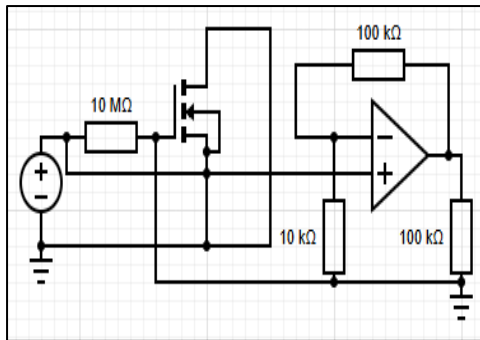
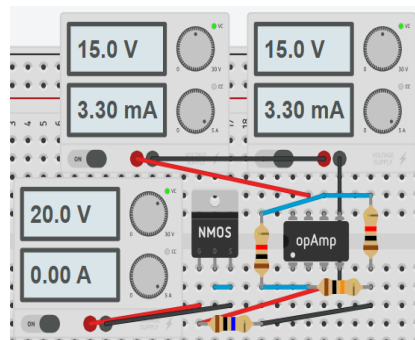
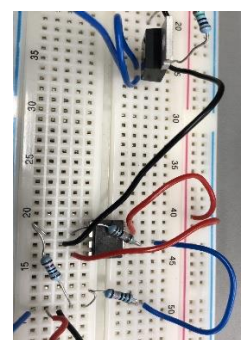


Figure 25: Ig Unity Amplified Model

The sensor is now connected to a resistor in series with the gate, supplied with 20V so the MOSFET is in on state. In this case the shunt resistor in series with the drain leakage current is kept the same at 10 M ohms. The drain and source are shorted. The DMM is used to measure the outputted resistor voltage shown in *table 10*. Initially a unity gain amplifier configuration is used then a gain of 11 and 101 is used to amplify the resistor voltage in series with the gate terminal.

When a unity gain, amplifier is used to measure the voltage across the gate resistor it seems that the voltage across the amplified output resistors gives a signal voltage unlike without the buffer. This can be due to the buffer outputting the signal voltage with a large impedance, however the value is too small to draw relevant conclusions relative to the DMMs absolute uncertainty of measurement. Regardless the gate leakage current is calculated from the voltage by dividing by the gain then dividing by the shunt resistor value of 10×10^6 as shown in *table 10*.

Figure 15: I_g Amplified DiagramFigure 27: I_g Amplified ModelFigure 28: I_g Amplified

When the linear gain of the amplifier is increased it seems to 11 and 101, the outputted amplified voltage is approximately a factor of 10 larger. When the respective gate leakage currents are calculated by dividing by the gain of also increasing factor of about 10. When dividing by the shunt resistor, all gate leakage currents fall within the nanoamp scale. This consistency indicates that the gate leaked current falls within this range. It seems that increasing the more accurately measures the gate leakage voltage due the values tending to a true value. A gain of 1001 was not used due to an erroneous measurement when used.

Amplification	1	11	101
Output voltage	0.007	0.099	0.85
Gate leakage I	0.7nA	0.9nA	0.842nA
Vgs	19.44V	17.69V	18.83V

Table 10: Amplified Gate Leakage Voltage

3.10 Measuring Gate Leakage Current with Sensor and Microcontroller

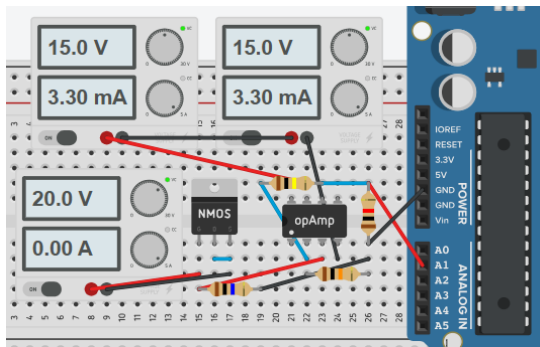


Figure 26: I_g Measured with Arduino Model

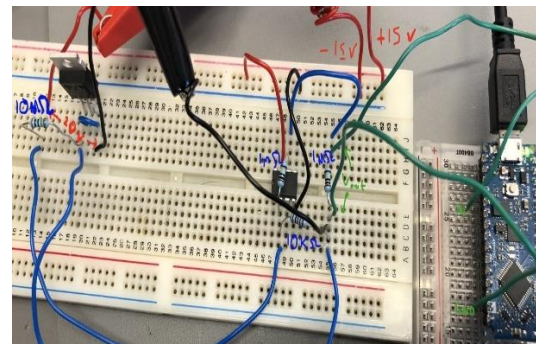


Figure 27: I_g Measured with Arduino Breadboard

The setup for the gate leakage current of the MOSFET with the sensor with amplification gain of 101 for accuracy. The Arduino microcontroller voltage sensor was implanted from previously with one side of the resistors electrical potential connected to analogue pin one and the other side connected to ground. The Arduino nano was used in this experiment due to its compatibility with the board used in the final design. This Arduino code was updated to calculate the mean value of voltage every 100 readings due to fluctuation in the voltage value due to voltage noise.

```
void setup() {
    Serial.begin(9600); // Start serial communication at 9600 baud
}

void loop() {
    const int numReadings = 300; // Number of readings to average
    int sumAnalog = 0;           // Sum of analog readings
    float sumVoltage = 0.0;      // Sum of voltage readings

    for (int i = 0; i < numReadings; i++) {
        int analogValue = analogRead(A1); // Read the analog value from A1
        float voltage = analogValue * (5.0 / 1023.0); // Convert to voltage (assuming 5V reference)

        sumAnalog += analogValue;
        sumVoltage += voltage;

        delay(5); // Delay between readings (500ms)
    }

    // Calculate the mean values
    float avgAnalog = sumAnalog / float(numReadings);
    float avgVoltage = sumVoltage / float(numReadings);

    // Output the mean values
    Serial.print("Mean Analog Value: ");
    Serial.print(avgAnalog);
    Serial.print(" | Mean Voltage: ");
    Serial.print(avgVoltage, 3); // Print voltage with 3 decimal places
    Serial.println(" V");

    // Small delay before next set of readings
    delay(10);
}
```

The output voltage is shown on the serial monitor on the Arduino IDE. Similarly, the DMM is also used to measure the output voltage with to compare and validate the results. Initially using the nxp op629 MOSFET, the DMM was used to measure the output amplified resistor. Like the Arduino IDE the output voltage measured voltage with a high frequency of fluctuation. The measured voltage fluctuated from 0.795V to 1.345V. The previous measured outputted voltages do fall within this range however gate leakage current cannot be calculated

form this unstable value. The reason for the fluctuations can be due to the extra impedance of the microcontroller as a load. Regardless the Microcontroller displayed an outputted voltage of 1.08V with a fluctuation of ± 0.02 , this is due to the mean calculation of the digital analogue read values in the IDE code. The Gate leakage voltage is then calculated by dividing by the liner voltage gain and the shunt resistor value.

$$I_G = \frac{V_{out}}{A_v * R_{shunt}} = \frac{1.08}{101 * (10 * 10^6)} = 1.069 \text{ nA} \quad (3)$$

To verify the accuracy of the calculation the percentage uncertainty of the gate leakage current is determined. Given precision resistors were used in this experiment which have a percentage uncertainty of manufacture value of 1%, the gain percentage uncertainty is determined from equation 2 which uses the quotient of tow resistors hence a percentage uncertainty of 2%. By finding the summation of each variable percentage uncertainty in equation 3. Percentage uncertainty is determined under 5% the calculation can be validated

$$U_{\%}(I_G) = 2\% + 1\% + \frac{0.02}{1.08} * 100 = 4.85\% \quad (4)$$

The validate the MOSFET gate leakage current calculation the nxp MOSFET is replaced with the IRL530 MOSFET and the outputted voltage of the sensor is measured with the DMM and the Arduino IDE. The DMM similarly measured a fluctuated voltage from 1.455V to 2.955V. Given this range is too large to calculate the gate leakage current, the microcontroller output a value of 2.055V in the IDE using the running mean code. Due to the fluctuations of the IDE outputted value there was a greater unitarity of ± 0.12 . The Gate Leakage voltage is then calculated using the same method.

$$I_G = \frac{V_{out}}{A_v * R_{shunt}} = \frac{2.055}{101 * (10 * 10^6)} = 2.035 \text{ nA}$$

The percentage uncertainty was then calculated to also validate the calculation using the known absolute uncertainties by calculating the summation of the individual variables in the gate leakage current equation. The calculated percentage uncertainty is greater then 5% suggesting the calculation is invalid however it is noted that there was a significant fluctuation in voltage measured by the DMM. It is also noted that there is an approximate 90% difference in the gate leakage current values so regardless of the significant percentage uncertainty the difference in gate leakage current values can be regarded to the difference in transistors not the uncertainty in measurements.

$$U_{\%}(I_G) = 2\% + 1\% + \frac{0.12}{2.055} * 100 = 8.84\%$$

3.11 LCD display

An LCD display compatible with 5V was purchased to be able to display the Arduino IDE outputted voltage values and calculated gate leakage current for the final circuit. The LCD display has 16 pins so an I2C module connected to the LCD display minimises the analogue control pins required to display an output on the display. To write code to control the LCD two libraries are required to digitally write on the display, one is <Wire.h> which is already installed on the used IDE 2.3.4 and <LiquidCrystal_I2C.h> which was installed from GitHub (Rickman, J., 2021).

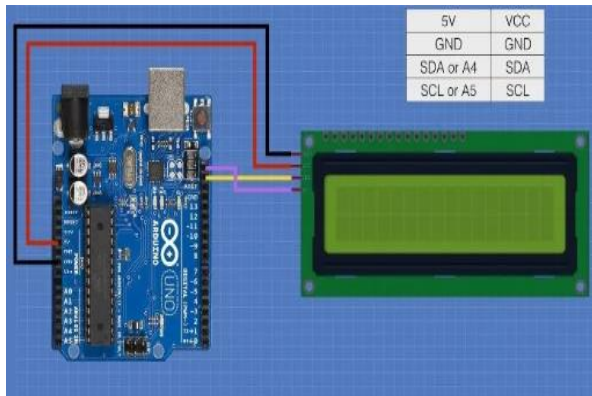


Figure 28: I2C LCD Configuration (Workshop, D. ,2018).

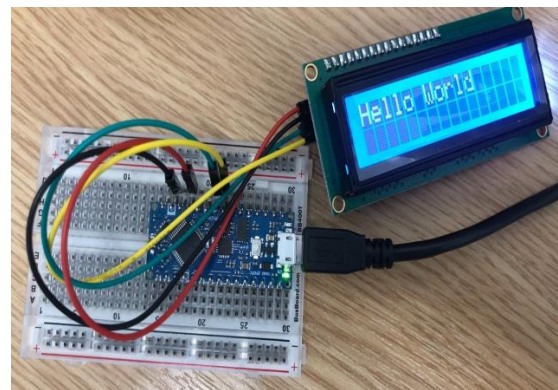


Figure 29: I2C LCD Arduino nano

The I2C module related to the Vcc connected to the 5V pin on the Arduino nano, The GND pin to GND on the Arduino, SCA representing the clock signal was connected to A5 and SDA representing the digital signals was connected to A4 pin on the Arduino nano. Before writing the code to display IDE output, the address of the LCD display needed to be identified. The address is defined by the combination of the of pins on back of the module for when using multiple LCD displays. The following code is used to determine the address in hexadecimal which outputted 1 device was found with address 0x20 in the serial monitor of the IDE. (Workshop, D. ,2018)

```
#include <Wire.h>

void setup() {
  Serial.begin (9600);

  // Leonardo: wait for serial port to connect
  while (!Serial)
  {
  }

  Serial.println ();
  Serial.println ("I2C scanner. Scanning ...");
  byte count = 0;

  Wire.begin();
  for (byte i = 8; i < 120; i++)
  {
    Wire.beginTransmission (i);
    if (Wire.endTransmission () == 0)
    {
      Serial.print ("Found address: ");
```

```

    Serial.print (i, DEC);
    Serial.print (" (0x");
    Serial.print (i, HEX);
    Serial.println (");");
    count++;
    delay (1); // maybe unneeded?
  } // end of good response
} // end of for loop
Serial.println ("Done.");
Serial.print ("Found ");
Serial.print (count, DEC);
Serial.println (" device(s).");
} // end of setup

void loop() {}

```

The LCD display configured to digitally write was connected to the microcontroller gate leakage current sensor as seen in *figure 30* and *figure 31*. The Arduino IDE code was changed now to output the gate leakage current value through using equation (3) with a gain of 101 implemented. The I2C display was initialised at the start through the command ‘LiquidCrystal_I2C lcd (0x20, 16, 2);’ using the determined address and known dimensions of LCD display form the datasheet. The output sensor voltage and gate leakage current was outputted on the LCD display and the IDE serial monitor, which both exhibited the same values.

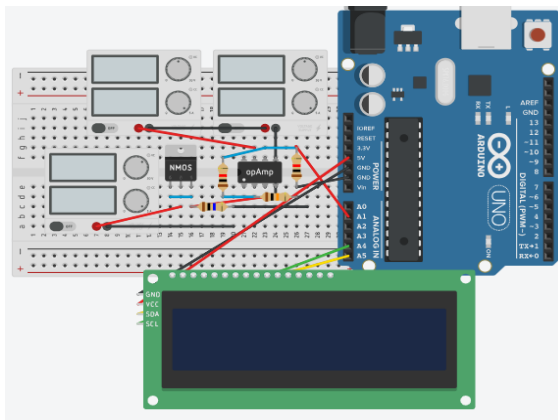


Figure 30: LCD displaying I_g Model

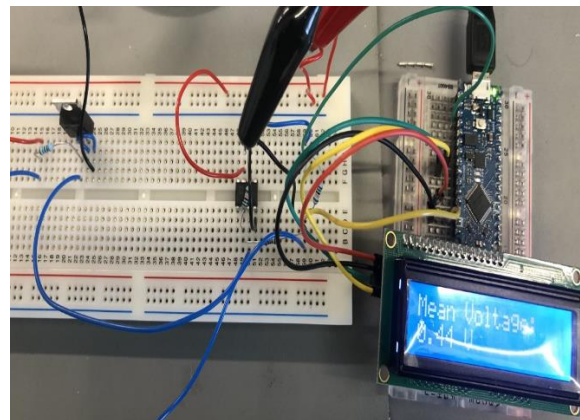


Figure 31: LCD displaying I_g

```

#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Initialize LCD with I2C address 0x20, 16 columns, and 2 rows
LiquidCrystal_I2C lcd(0x20, 16, 2);

#define NUM_SAMPLES 50 // Number of samples for averaging

void setup() {
  Serial.begin(9600); // Start serial communication
  lcd.init();         // Initialize the LCD
  lcd.backlight();    // Turn on the backlight

  lcd.setCursor(0, 0);
  lcd.print("Voltage & Current");
}

void loop() {

```

```

float voltageSum = 0.0;

// Read 50 samples and calculate the mean voltage
for (int i = 0; i < NUM_SAMPLES; i++) {
    int sensorValue = analogRead(A0);
    float voltage = sensorValue * (5.0 / 1023.0); // Convert to voltage (assuming 5V reference)
    voltageSum += voltage;
    delay(5); // Small delay between readings for stability
}

float voltageMean = voltageSum / NUM_SAMPLES; // Average voltage
float currentNanoAmps = (voltageMean / (101.0 * pow(10, 6))) * pow(10, 9); // Convert Ig to
nanoAmps (nA)

// Display on Serial Monitor
Serial.print("Mean Voltage: ");
Serial.print(voltageMean, 3);
Serial.print(" V, Mean Ig: ");
Serial.print(currentNanoAmps, 3);
Serial.println(" nA");

// Display on LCD
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("V: ");
lcd.print(voltageMean, 2);
lcd.print(" V");

lcd.setCursor(0, 1);
lcd.print("Ig: ");
lcd.print(currentNanoAmps, 2);
lcd.print(" nA");

delay(1000); // Update every second
}

```

4 Design Process

4.1 Schematic

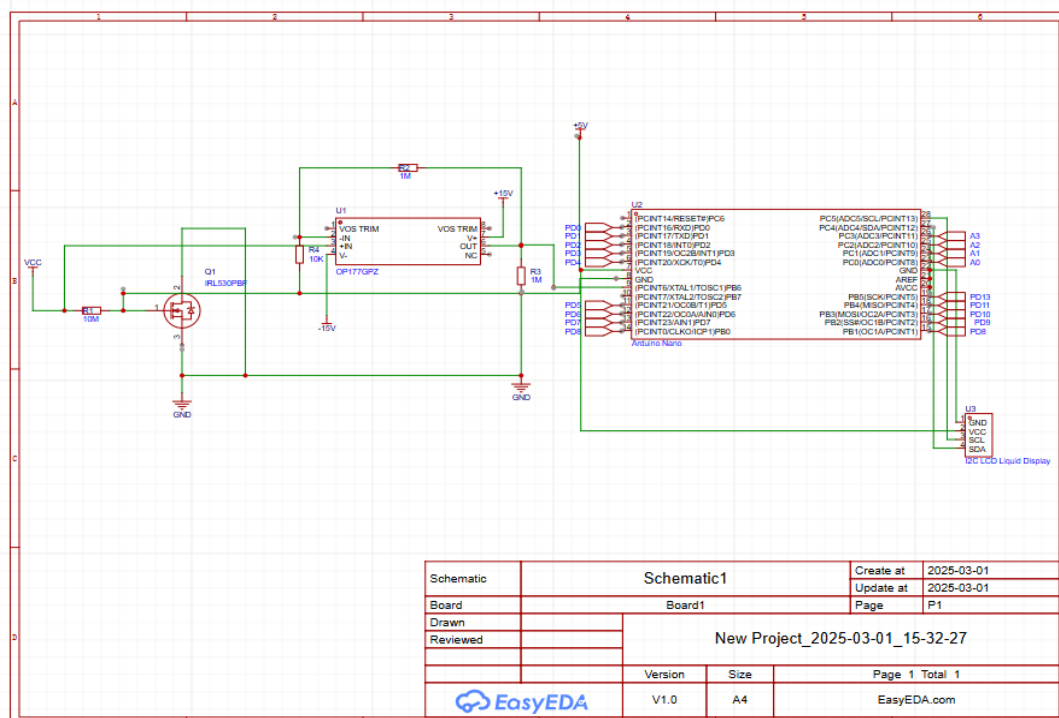


Figure 32: Circuit Schematic

Once the circuit was built on the bread board the next step in the project was the build the circuit using a PCD to be used for various MOSFETs. For this to be achieved a transistor holder was purchased to able to input different MOSFETs into the circuit. A schematic first drawn to be able to visualise the layout of the components before soldering in the PCB strip board. Unlike the individual designed models and circuit schematics created in the experimentation process all the components were set in a single frame to ensure the circuit configuration is coherent and the ground is relative to the entire circuit is determined.

Given the microcontroller uses five pins, with the ground pin used for both the I2C LCD display and the ground of the output sensor resistor. This was kept in mind during the physical assembling process.

4.2 Transistor Holder PCBs

All the electronic components purchased and used form the engineering electronic labs fit on the strip board despite the transistor holder and the banana plug holders, which do not align with the distribution of the pin holes on the bread board or the strip board. To compensate for this PCBs where designed and build to be able to hold the component in the boards without need to be concerned about faulty or dodgy connections.

PCBs were designed in Altium, using the schematics 2D and 3D dimensional schematics. The PCB where then processed by via emailing the PCB designs to the University of Warwick Electrical and Electronic Support Group Office. These prints took about a week to process, which need to be taken to account when focusing on the time schedule of the project.

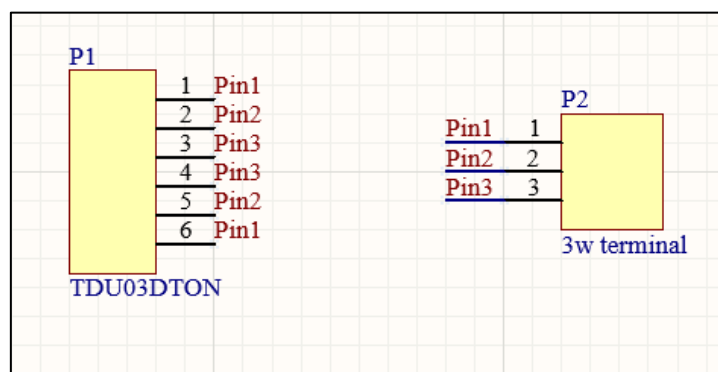


Figure 33: Transistor Holder PCB Schematic

The following figures *figure 34* *figure 35* and *figure 36* are representations of the transistor holder. Digi Key Sullins Connector Solutions transistor holder (DigiKey Electronics.,2025). The holder has 6 pins which are in pairs of 3 which do not align. The reason for this is not understood, regardless the PCB shorts the adject pins creating a connection by conduction. Each adjacent pair of pins correlate to Gate, Drain and Source pins which all align to an individual signal pin. These pins are equivocal in distance which aligns with the distances between pin holes in a standard bread board.

The three pins were soldered into the PCB strip board in the final design. The DMM was used to measure the resistance between the pins of the MOSFET and between the pins on the PCB to see if they align to validate the conductivity on the PCB

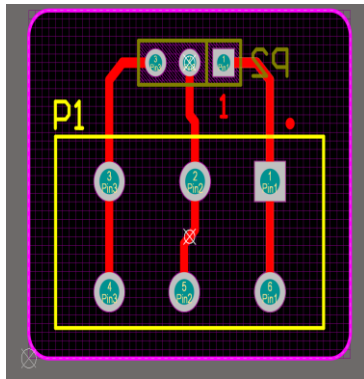


Figure 34: Holder PCB 2D View

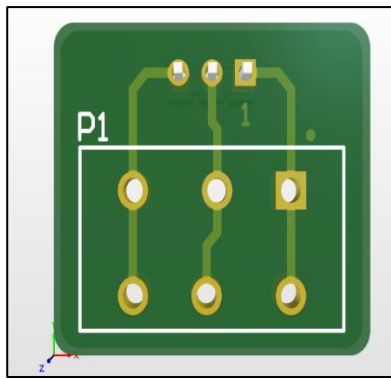


Figure 35: Holder PCB 3d view

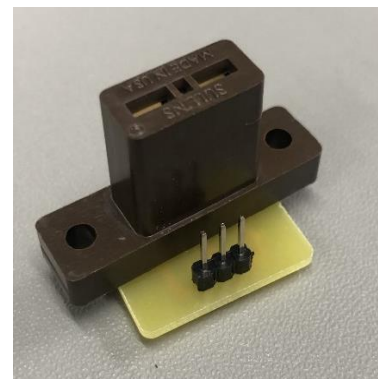


Figure 36: Transistor Holder PCB

4.3 Power Banana Cable holder PCBs

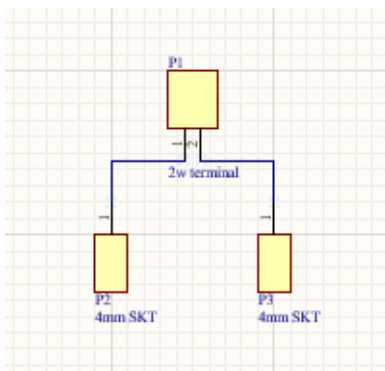


Figure 37: Power Cable holder PCD Schematic

Using Altium The PCD for the power cable sockets were designed. The schematic simply represents how the copper metal tracks connect via conduction, simply 2 pins holes for the sockets which were connected to 2 individual pins used to solder into the strip board to supply voltage and act as a relative ground. The sockets were provided by The University of Warwick Electrical and Electronics Support Group (EESG).

The following figures *figure 38*, *figure 39* and *figure 40* shows a visual representation of the PCBs produced for the final circuit. The banana sockets were placed in Altium to be able to understand the pin size and spacing. It was important to design the PCBs as simple as possible to prevent voltage being lost from the connections. It was also important to keep the positive and terminal connections electrically isolated from each other to prevent a short circuit removing the applied potential differences. This PCB was built three times for three PSU channel inputs. Given a positive and negative power supply is required for the op amp the series and ground configuration two PSU channels was configured after the sockets on the board.

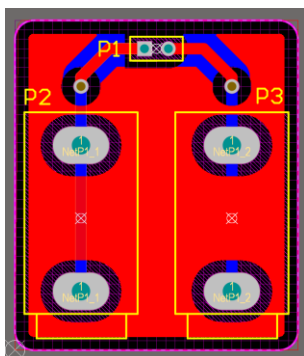


Figure 38: Power Socket PCD 2D view

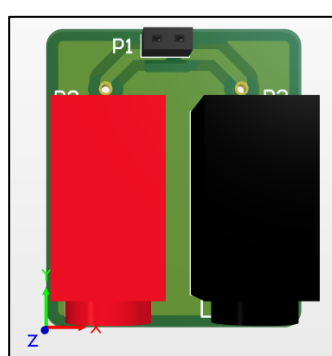


Figure 39: Power Socket PCD 3D view

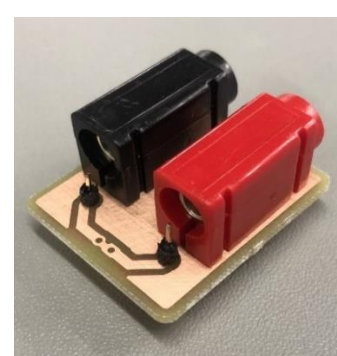


Figure 40: Power Socket PCD

4.4 Final Circuit

When all the components were delivered to the University of Warwick Engineering Department and the PCBs were processed and built, the final circuit building process began. To give sufficient time for the building process and taking results, the ordering and components and processing of PCD requests were scheduled and organised in the supporting logbook due to the timing relative to the project deadline.

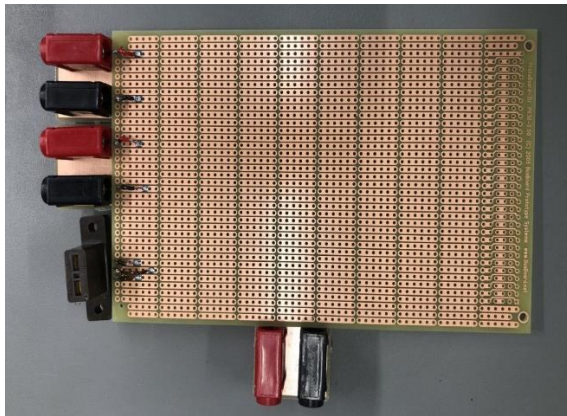


Figure 42: Strip Board with PCDs Soldered

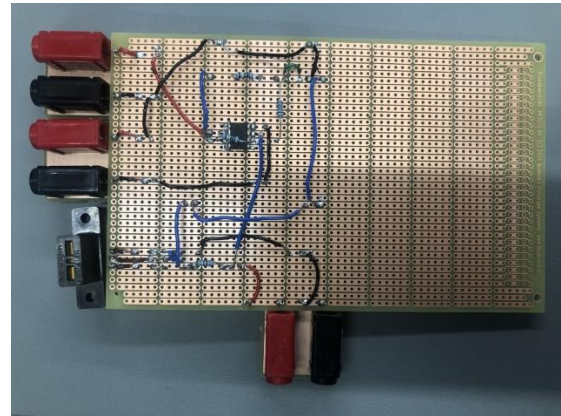


Figure 42: Strip Board with sensor

During the production process of the final board, the DMM was used to verify the validity of the conductive connection. The First step was to connect the built PCBs to the sides of the board, when the sockets were connected to the PSU the DMM was used to measure the track voltages to verify the connections. This was also done after the sensor was built to measure the output voltage of the sensors output resistor. The circuit configuration and code were the same as the final experiments bread board model.

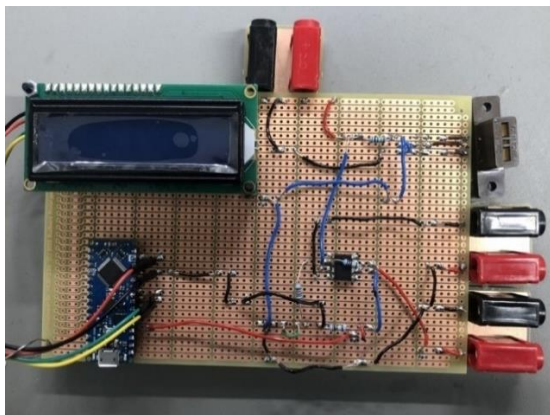


Figure 43: Final Circuit no power

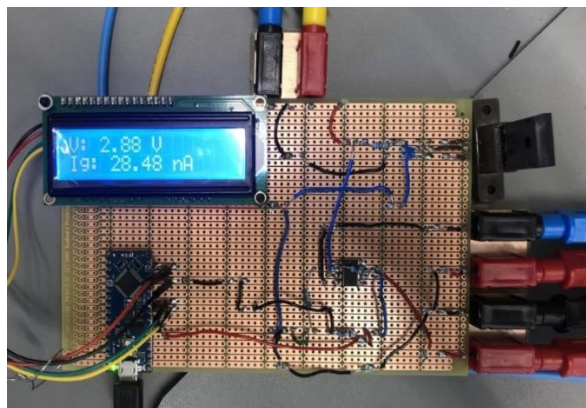


Figure 44: Final Circuit with power

5.1 Results

5.2 Measured Values

The gate leakage current was already measured, during the experimentation process using the smaller 10 mm width transistors, IRL 530 MOSFET and the nxp op629 MOSFET. Now the designed circuit was used to measure two CREE manufactured MOSFETs, (Cree LED, 2024). to test the reliability of the circuit to use for a range of transistors.

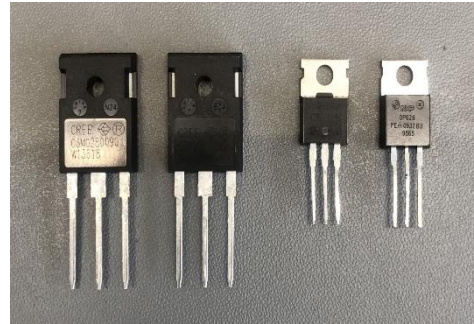


Figure 45: Sample measured MOSFETs

First a CREE MOSFET manufacturing number: C3M0280090W14218 J was measured. From the data sheet the silicon carbide MOSFET has a typical gate leakage current of 10nA with a maximum value of 250nA. This is measured under the conditions that the gate source voltage (V_{gs}) is 15V and the drain source voltage (V_{ds}) is 0V. (alldatasheet.com, 2021). From the circuit the MOSFET was inserted with the connected polarity into the transistor holder and the gate source voltage input with 15V from the PSU.

The output measured voltage and gate leakage current was 3.16V and 28.7nA respectively which was output on the serial monitor of the Arduino IDE and the LCD display. Given analogue read value is between 0V and 5V and the gate leakage voltage is around 10nA and below stated maximum value on the datasheet. The gate leakage current scale can be validated from the data sheet of the MOSFET however the exact value cannot be validated from this measurement. It is also noted that the voltage value fluctuated by $\pm 3.50V$. The gate leakage current was calculated from the output sensor resistors voltage from equation (3):

$$I_G = \frac{V_{out}}{A_v * R_{shunt}} = \frac{3.16}{11 * (10 * 10^6)} = 28.73 \text{ nA}$$

The gate leakage current calculation validates that the code correctly interprets the current measurement from the read analogue value. This calculation cannot validate the accuracy of the measured value, but the precision can be determined from the percentage uncertainty, with the same method used in equation (4):

$$U_{\%}(I_G) = 2\% + 1\% + \frac{3.50}{28.7} * 100 = 15.18\%$$

The percentage uncertainty is slightly high as it is above 5%. This is due to the large amount of voltage noise measured by the analogue read on the microcontroller. This can be due to there being a lower output impedance in the configured circuit. Regardless of the precession of the results the range of the accuracy can be validated but not the exact value.

Another CREE MOSFET was tested with manufacturing number C3M0280090W13618 D. The datasheet of the MOSFET also stated that a typical gate leakage current is 10nA with a maximum value of 250nA. This is measured under the conditions that the gate source voltage (V_{gs}) is 15V and the drain source voltage (V_{ds}) is 0V. (alldatasheet.com, 2021). The circuit was thus set up the same with the MOSFET inserted with the connected polarity in the transistor holder and the gate source voltage input with 15V from the PSU.

The output measured voltage and gate leakage current was 0.38V and 3.5nA respectively displayed on the serial monitor of the Arduino IDE and the LCD display. This is within the range of around 10nA, so the range of the value is valid. It was stated by the project supervisor that one of the two transistors had a faulty gate. Given that the gate leakage current is below the typical value on the data sheet it can be concluded that this is the faulty transistor, given the value is not a trace reading the gate leakage current can be below the gate leakage current due to ageing effects wearing the insulation in the MOSFETs gate. It is also noted that the voltage value fluctuated by $\pm 0.07V$. The gate leakage current was calculated from the output sensor resistors voltage from equation (3):

$$I_G = \frac{V_{out}}{A_v * R_{shunt}} = \frac{0.38}{11 * (10 * 10^6)} = 3.45 \text{ nA}$$

The gate leakage current equation validates the code correctly interprets the gate leakage current from the analogue read value. Even though the MOSFET has a lower gate leakage current due to a faulty gate, the true value of gate leakage current is now known and thus the precision cannot be varied.

$$U_{\%}(I_G) = 2\% + 1\% + \frac{0.07}{0.38} * 100 = 21.42\%$$

The calculated percentage uncertainty is significantly larger than 5% making it difficult to validate the calculated precision. The uncertainty is large due to the voltage measurement being very small in proportion to the range of 5V readable by the Arduino and with the same voltage noise proportional this range with a smaller output voltage leads to a larger percentage uncertainty of gate leakage current.

5.3 Conclusion

MOSFET	IRL 530	nxp op629	C3M0280090 J	C3M0280090 D
I_G	1.069nA	2.035nA	28.73nA	3.45nA
$U_{\%}(I_G)$	4.85%	8.84%	15.18%	21.42%

Table 11: All MOSFETs Measured Gate Leakage Currents

To conclude the project, the gate leakage current was successfully measured within the nanoamp ranges stated on respective MOSFET datasheets. The calculated percentage uncertainty verifies measured precision with larger uncertainties justified. The final circuit shows a difference in measured gate leakage current measurements successfully identifying a faulty transistor gate.

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